

OFF THE GRID

**WATER HARVESTING,
TREATMENT,
AND STORAGE**
2nd Edition



MICHAEL CUSHING

Off The Grid: Water Harvesting, Treatment, and Storage

2nd Edition



Michael Cushing

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Table of Contents

[Copyright](#)

[Table of Contents](#)

[Introduction](#)

[Chapter One](#)

[Off The Grid Living, Water, and You](#)

[The Demands of Sourcing Your Own Water Reserve](#)

[The Dangers of Contaminated Water](#)

[Viruses](#)

[Bacteria](#)

[Parasites](#)

[Microscopic Plants](#)

[Chapter Two](#)

[Water Collection](#)

[The Water Demands of Off the Grid Living](#)

[The Water Cycle](#)

[Worldwide Water Distribution](#)

[Finding a Source of Water for Your Homestead](#)

[Identify potential water sources in your chosen location](#)

[Test samples from your potential water sources](#)

[Primary Water Sources](#)

[1. Rainwater](#)

[2. Groundwater](#)

[3. Surface Water](#)

[Harvesting Water](#)

[1. Rain Collection](#)

[The Three Classifications of Roof Collection Systems](#)

[Non-Potable Water for Outdoor Use](#)

[Non-Potable Water for Indoor Use](#)

[2. The Well](#)

[3. Springs and Other Natural Surface Water Sources](#)

[4. Dew Collection](#)

[Chapter Three](#)

[Water Treatment](#)

[Making a Water Filter](#)

[Materials](#)

[Steps in Constructing a Water Filter](#)

[Methods of Using Bleach to Purify Drinking Water](#)

[Purifying Water with Iodine](#)

[Potassium Alum for Water Purification](#)

[Using Natural Products](#)

[A Solar Still](#)

[Solar Disinfection of Drinking Water](#)

[Solar Pasteurization of Water](#)

[Using Heat](#)

[Chapter Four](#)

[Storing Your Water](#)

[Types of Water Containers](#)

[Calculating Your Storage Requirement](#)

[Storage Tank Sizing with Water Balance Tables](#)

[Tips for Keeping Your Stored Water Clean and Fresh](#)

[Conclusion](#)

[Other Related Books](#)

[The Water Collection Handbook: A Guide To Collecting, Treating, and Storing Water](#)

[Off the Grid 101: An Introduction to Living A Self-Sustaining Lifestyle \(2nd Edition\)](#)

[Photo Credits](#)

[Bonus](#)

Introduction

Thank you for purchasing this book. Within these pages you will learn how to store, treat and collect water using a variety of techniques and methods that are suitable for those living "Off the Grid" or without access to a municipal water supply. Whether you are seeking to start your own independent homestead and need to provide water for your personal use, livestock and gardening needs, or just want to set up a few rain barrels to cut down on the water bill this book has you covered.

Saying that water is essential is an understatement. After being able to breathe, it is water that makes life on this planet possible. Many people have the luxury of not worrying too much about where their water comes from or whether or not it is safe for human and animal consumption. They just turn on the faucet and there you go! Suitable water for man and beast alike.

The truth about water is far more interesting than it would seem to the casual observer. Even though water covers a majority of the Earth's surface, about two-thirds or sixty-six percent of it is unfit for human use. Much of it is too salty or contains high levels of dangerous contaminants. After everything is said and done, only about three percent of the Earth's water is considered safe enough to drink without some sort of processing or treatment to make it fit for human use.

Those living on the grid don't have to worry about how their water becomes pure or where it originated from. These issues are generally of no concern to the average, ordinary person, living an average normal lifestyle.

Those choosing to live off the grid often don't have the advantage of a municipal water supply to meet their needs. There is no faucet connected to an endless supply of fresh water. Water, like everything else in the homesteading lifestyle, is the sole responsibility of the homesteader. It is you who must locate and maintain access to a water supply, make sure that it is safe and find a way to store it for future use.

This is no easy task. It is not impossible, though. In this book, we'll explore the options available for those wanting to establish a safe and reliable way to supply water to their homestead. You will learn about the common ways water can become contaminated, see what well-digging entails and how to collect rainwater. We will also take a look at a few ways to prepare your water for use and keep it safe.

If you're considering the off the grid lifestyle, providing yourself with water is an important priority and a task full of interesting challenges. It is my hope that this book will provide you with the information you need to confidently meet and rise above these and the other challenges you face as you establish your homestead.

Mike



Chapter One

Off The Grid Living, Water, and You

The off the grid lifestyle is not one that is suitable for everyone. It takes a certain type of individual to meet and overcome the challenges inherent in OTG living and homesteading. It's not an easy way to live. The daily demands made of the modern homesteader are intense and many.

When people decide to move away from the grid, they no longer have access to public utilities such as electricity and water service. It is also common, in this lifestyle, for a person to produce all of their own food. This commonly entails a combination of gardening and raising small livestock. Essentially, a person living off the grid is solely responsible for everything that they need to survive.

The individual must supply all food, shelter, water, electricity, waste removal, and other requirements in homesteading. Some of these needs are more easily met than others. A house, for instance, can be built without too much difficulty and has the added bonus of only needing to be done once. Shelter options such as mobile homes, make this requirement fairly easy to meet.

Electricity is a bigger puzzle to solve, but the innovative homesteader will be able to put a solution together. Many options for alternative energy exist providing specific choices to fit just about every homesteading situation. Windmills, solar power, and gas (or other type) generators are all viable options. Many OTG homesteads utilize a combination of these methods to provide consistent and adequate electrical service. While there is always a degree of uncertainty inherent in being responsible for your own electrical service, it isn't too much trouble to establish a fitting solution that can rival the public grid in terms of dependability and consistency.

The Demands of Sourcing Your Own Water Reserve

Unlike with traditional housing, off the grid-ers don't have the luxury of relying on a third party (in this case the municipal water company) to meet their need for clean drinking water. There is no source of easily accessible and pre-treated water available, it must be provided for by the individual. It also has to be renewable and sustainable. It must be clean and safe. A source of water is no good to you if it is in danger of drying up or if it is a breeding ground for harmful microorganisms.

Water differs from electricity in that it can't be produced simply by burning a fuel source or using batteries. It needs to be provided by natural mechanisms and then found and collected. It is not enough to just have access to water. Water for domestic purposes needs to be free of viruses, microorganisms, and chemical contaminants. After a source of water has been located, treatment and purification are the next steps to take to ensure its suitability for human and animal consumption. There must also be a way to store the treated water for an extended amount of time.

Just about everything that is required to run a homestead is dependent on having a renewable source of water. Aside from the water that is needed to sustain human life, there must be enough water for the crops, to feed the livestock, for washing the dishes and clothes, for waste removal, and all the other tasks that uncontaminated water is commonly associated with. Potential water sources should adequately meet your homestead requirements throughout your stay in an off-grid area.

The Dangers of Contaminated Water

Contaminated water can cause a host of health problems and may even result in death. Not only for you but for your livestock and crops as well. Bad water can decimate your herds and flocks and wreak havoc on your crops, leaving you at a considerable disadvantage when it comes to sustaining your OTG lifestyle.

Using untreated water from a potential source is never a recommended course of action. In fact, it is the number one ‘don’t’ of homesteading when it comes to water use. Dangerous water contaminants are classified into different categories:

Viruses

A virus is a microscopic semi-living agent that can only stay alive by infecting a living host and hijacking certain cellular mechanisms. A virus differs from bacteria in the sense that they are not complete organisms unto themselves, they very much need a host to live. A person or animal that has become infected with a virus will find no relief in medications such as antibiotics. A virus does what it does, and the host is either strong enough to withstand the rude house guest's intrusion or it perishes. Any virus is difficult to deal with; many are very resilient and can withstand high temperatures or lay dormant for years under wretched environmental conditions, just waiting for a host.

Medications are available for some types of viral infections, but they tend to focus on managing symptoms in an effort to ease the discomfort experienced while the virus runs its course, rather than target and destroy it. Boiling and chemical treatment methods are effective ways to cleanse your water of any viral threats.

Common waterborne viruses include some types of Swimmer's Ear (Otitis Externa), Typhoid, SARS, and Vibrio Illness.

Bacteria

Bacteria, unlike viruses, are living organisms unto themselves. There are countless types of bacteria, not all of them are harmful, and many are good to have around.

These tiny bacteria exist everywhere, especially in water. They thrive in bodies of water and quickly pass into human or animal hosts and can cause quite a number of illnesses. Legionnaire's disease, dysentery, E. coli infection, Botulism, and other gastrointestinal diseases are all common types of water-borne bacterial illnesses.

Thankfully, managing bacteria is usually a lot simpler than removing viruses. Many common types of bacteria can be dealt with through boiling and filtering the water before consuming or using it.

Parasites

Parasites come in many forms, unlike viruses and bacteria; they are multi-celled organisms - usually worms and similar insect-like creatures. A parasite infects a host, feeds on it and lays its eggs within it. The eggs hatch inside the host and continue to feed and reproduce, causing all manner of health problems in the process.

Microscopic Plants

This group of water contaminants includes fungus, algae, and other tiny plants collectively called "Protozoa." They thrive in water containing fecal matter and other bio waste. Once they enter a host, they can cause skin conditions and various illnesses.

Consuming contaminated water is only one way to put yourself at risk to the above dangers. It is just as likely that someone will become ill through using water to wash or clean yourself as they are through using it to cook with and drink.

Ensuring your water supply is clean and safe to use is the number one priority to the homesteader when it comes to water. It doesn't take much thinking to reach the conclusion that without a clean reliable source of water any dreams you had of living an independent, off the grid lifestyle quickly vanish in a haze of nightmarish parasitic infections.

Access to a sustainable source of water is something that is often taken for

granted by much of the population. You don't have that option when living outside of the traditional system.

Figuring out how to gather and care for your water supply should be high on the list of every prospective homesteader.

In the remainder of this book, we will be discussing several options for ensuring that you have a consistently safe manner to provide all of the water that you need to properly care for yourself and your land.

Chapter Two

Water Collection



Water has played a crucial role in our lives since the establishment of early civilizations. For 5,000 years, freshwater sources have been diverted to agricultural lands to grow crops and rear livestock. Currently, the agricultural, residential, commercial, and industrial sectors drive the increase in fresh water demand over the past years. While surface water sources primarily meet this demand, some localities depend on underground aquifers to supply their daily water requirements. Here are a few tidbits on the history of water use since the birth of the human race:

- About 12,000 years ago, hunting-gathering ethnic groups regularly

moved from one fertile river valley to the other to survive. Looking back, most civilizations started along riverbanks including the ancient Mesopotamian civilization.

- The reality of water shortages appeared more than 7,000 years ago and spurred people to develop irrigation systems, giving birth to the agricultural sector.
- Drought brought an end the Mayan civilization 1,100 years ago.
- During the mid-1800's, citizens of major North American towns, including Chicago suffered from typhoid fever and cholera due to fecal contamination of their primary surface water resources.
- The year 1858 became known as the *Year of Great Stink* in the English town London as sewage and waste accumulated in the Thames.
- From the latter part of the 19th century to the early parts of the 20th century, dams emerged as the most popular water management facility.
- The 20th century saw the rise of human dependence on water for agricultural use.
- Passage of the Clean Water Act in 1972 raised human awareness on the

issue of water contamination and the need to safeguard freshwater resources.

The importance of water to human survival is unparalleled. In fact, among all homestead requirements, water is the make-or-break resource. Some homesteaders are fortunate to have bountiful groundwater sources. Others who choose to settle in less populated, remote areas often experience an unreliable supply that has either high mineral content or is extremely difficult to find. While transporting water from nearby cities can be an option, it is a more expensive route to take. Furthermore, the risk of a vehicle breakdown in the middle of nowhere is present when you choose to live off-grid. Before plunging into your rural adventure, it is logical to have an overview of the practical details of sourcing your very own water supply.

The Water Demands of Off the Grid Living

The average water use among Americans is eighty to four hundred gallons a day, where 37.5% goes to toilet flushing (18%) and bathing (19.5%) as of 2010, according to the American Water Works Association (AWWA). This value is 6% lower than the 1999 figure published by the same organization (43.5% of residential water use). The use of efficient toilets and low flow showerheads largely contributes to this decrease in water demand.

Meanwhile, other indoor water use include: laundry (22.1%) and dishwashing (1.5%). Around 23.9% of indoor water use involves water flowing from faucets while 8.8% is lost to leaks.

In a homesteading situation, you may not be running a washing machine or hosing the car down every weekend, but you will still be using quite a bit of water, albeit lower than the average American household, throughout the day. You'll be watering your crops, and making sure any livestock has enough water. This is in addition to meeting your personal water needs. The bottom line is that life requires a lot of water. If you intend to be successful in your homesteading endeavors, you have to know how much water you will need so you can devise a plan to make sure that you can meet those needs.

Calculating your estimated water needs must take into account the size of your homestead, the number of humans and animals who are living there and any agricultural projects you are considering. A simple vegetable garden uses considerably less water than a large hydroponic or aquaponic setup; goats need more water than chickens, etc.

Correctly estimating your water requirements will save you many headaches, and it is an important first step in devising a water plan. Make sure you take your time and double-check your numbers to avoid grief down the line.

The U.S. Geological Survey (USGS) gives a rough estimate of how much water goes into the daily activities of each person living in your home:

Bathing in a tub:	36 gallons
Shower:	5 gallons per minute (just multiply how much time it takes for each person to shower, factor in the number of times each person takes a bath per day and you'll get the total daily water volume)
Brushing your teeth:	less than a gallon
Washing your hands and face:	1 gallon
Shaving your legs or face:	1 gallon
Using the dishwasher:	around 6 to 16 gallons
Manual dishwashing:	around 8 to 27 gallons
Clothes washing:	25 gallons
Toilet flushing:	3 gallons
Drinking water:	8 ounces per average sized glass
Outdoor water use:	2 gallons per minute on the average; this also depends on the

size of the lawn or agricultural area and the pressure from the faucet. A quarter-acre garden needs up to 350 gallons on a summer day, while livestock can use up to 100 gallons of water.

Based on these figures you can calculate your average homestead water requirement per day. Of course, you need to factor in the weather conditions and the available local water reserves in your area. Is the weather condition generally dry throughout the year? Do you have quick access to groundwater or surface water (rivers, lakes, springs)? A safe estimate of your annual water needs with consideration of dry months is to multiply your average daily use by 365 (total number of days in a year) plus 90 days.

~~*Total Annual Water Requirement = Average Daily Requirement (365 days in a year + 90 days)*~~

If your homestead requires 500 gallons of water daily, your annual water requirement would be:

~~*Total Annual Water Requirement = 500 gallons per day (365 days in a year + 90 days)
= 227,500 gallons*~~

While it can be rare for residents of temperate areas to experience a three-month dry spell, this is typical for people living in central California or in the southern part of Oregon. If you live in areas that experience longer dry seasons, change the factor to 180 days (6 months) or adjust based on the typical length of the hotter months.

While living on the grid, it is easy to take for granted just how easy it is to access clean water from a reliable source. This is a luxury that many homesteaders and off the grid enthusiasts don't have.

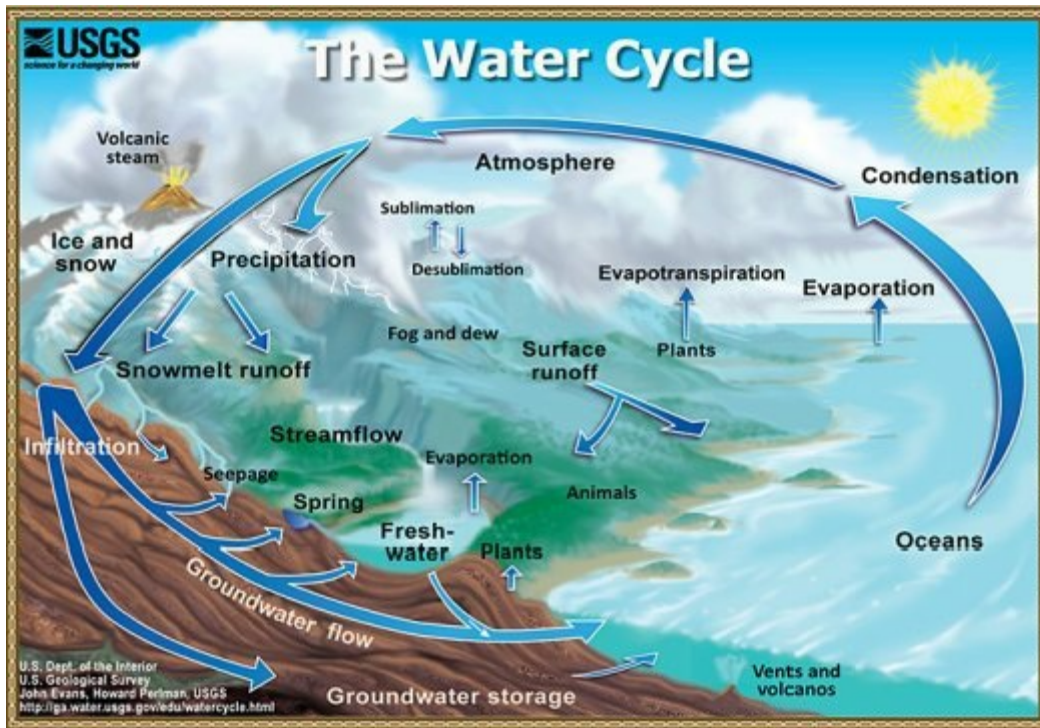
Homesteading often means that you're on your own and responsible for all that befalls you. When it comes to collecting water, it is a good idea to have at least three sources at your disposal. Having access to several sources and the knowledge of various water harvesting techniques ensure that your water needs are adequately met. Relying on one source of water is not recommended because there is always the possibility of something going wrong - wells and springs can and do run dry and water sources can become contaminated with chemical waste or coliform. It is good policy to know more than one way of gathering water. But first, it's time to take a step back and see how water is transported and stored within the Earth.

The Water Cycle

One of the natural phenomena that sustain life on Earth is the water cycle. As the sun's rays hit the surface of the Earth, water bodies are heated up causing some of their liquid particles to evaporate, transforming into the gaseous water vapor. In the Polar Regions where most of the glaciers are located, the ice directly converts to water vapor albeit at a slower rate. Rising warm air currents bring the water vapor higher up in the atmosphere until it reaches the cooler regions and condenses to form clouds. Movements of air currents across the globe cause clouds to collide and become dense until the water vapor particles fall out of the sky and return to the ground in the form of raindrops, snow, or hail. Here's a fun fact: Did you know that when snow accumulates, frozen water is stored for millions of years in the form of ice caps and large glaciers?

While most of the water that precipitates returns to the ocean, a portion fills up surface water reservoirs such as rivers and lakes. Another part seeps into the ground to replenish aquifers, which store large amounts of freshwater, for a considerable amount of time. In some cases, the water finds its way into tiny openings in the land surface. As soon as the voids are filled, the water builds up sufficient pressure to break the existing land barrier and emerge as springs. The rest of the water particles become absorbed by plants.

Throughout the ages, water moves in this manner, transforming from one form to another (solid – ice, liquid – water, gas – water vapor) to sustain all living beings.



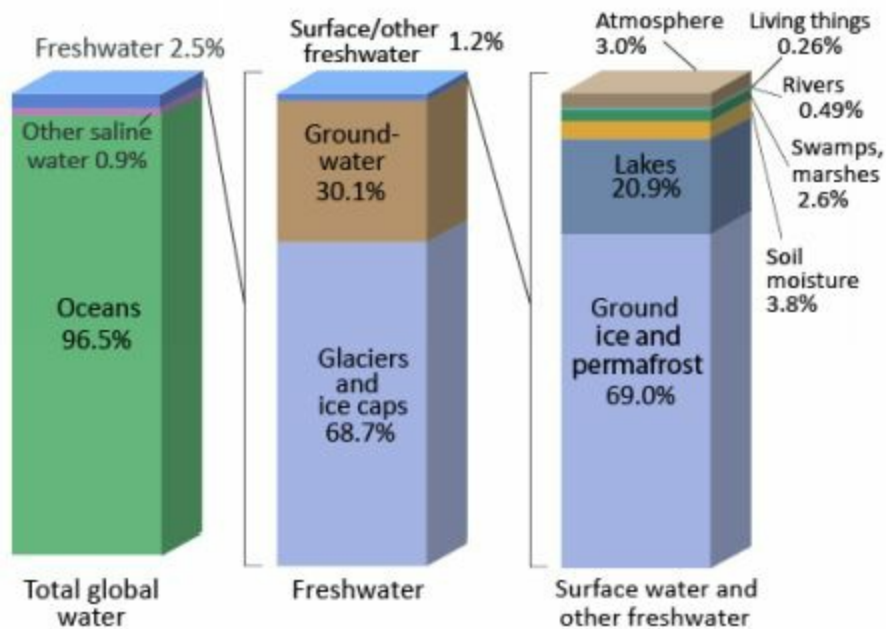
Source: <http://water.usgs.gov/edu/watercyclesummary.html>

Worldwide Water Distribution

About 70% of the Earth's surface is covered with water. This is equivalent to 332 million cubic miles of this life-giving fluid. Oceans are the largest water reservoirs containing 96% of the global water supply. Meanwhile, total freshwater that accounts for 2.5% of the global supply is stored in glaciers (68.7%), surface water (1.2%), and groundwater (30.1%).

While saline water is most abundant, its direct use for agricultural, industrial, and domestic purposes is limited. In most cases, it has to be processed to remove its salt content and reduce its mineral content to make it fit for consumption. Ironically, freshwater, though limited (2.5% of the global water supply), matters more for the daily survival of the world's total population.

Where is Earth's Water?



Source: Igor Shiklomanov's chapter "World fresh water resources" in Peter H. Gleick (editor), 1993, *Water in Crisis: A Guide to the World's Fresh Water Resources*.
NOTE: Numbers are rounded, so percent summations may not add to 100.

Various regions around the world receive varying distributions of surface water and groundwater. According to the United Nations World Water Development Report 2, in areas with a temperate climate, 33% of the total precipitation recharges groundwater reserves. Another 33% becomes surface runoff while the remaining 33% evaporates into the atmosphere as part of the water cycle. For regions with semi-arid climates, only 20% replenishes the groundwater reserve while 30% becomes surface runoff. The remaining 50% evaporates into the atmosphere. Arid regions approximately receive up to 200 millimeters of precipitation annually (about 13% of the 1,500 mm received by temperate regions). From this amount, only 1 percent goes to the groundwater reserves while 29% becomes surface runoff. It is no surprise that 70% evaporates into the atmosphere given the high temperature of the

immediate environment.

Given this information, you can adequately prepare your homestead water needs wherever you are located. People residing in drier regions may have limited water reserves underground and yet find abundant supply in nearby surface water sources. Furthermore, knowledge of the water cycle gives you insight on how you should store your water. People residing in arid regions will experience scarcity of water more often than those who live in temperate regions and thus may consider larger cisterns or storage facilities to have adequate water supply.

Finding a Source of Water for Your Homestead

An understanding of your landscape features and knowledge of the water cycle will give you ideas on where to find fresh water. There are various sources for drinking, garden irrigation, and livestock. Some may require extensive purification before use, such as exposure to ultraviolet light treatment while some can be directly used as it is. Some systems utilize chlorine for water treatment typically followed by charcoal filters to recapture the chlorine in the water. In any case, caution must always be taken when choosing potential water sources to ensure cleanliness and safety. Methods for channeling these water sources into your home can be as simple as digging a canal that leads the water towards your storage facility (cistern, tank, etc.) to installing high-pressure pumps to lift water from the deep wells.

Identify potential water sources in your chosen location

When you decide to live off the grid, the first step to take is to survey your property prospects before deciding to buy. One of the things you need to check is the available water reserves in the area. Is it near a lake, river, or spring? Is there an aquifer below the property? If there is, how deep is the water table? Consider obtaining an honest

quotation from reliable and reputable companies to have some idea how much and how long will it take to develop a sustainable water source.

Test samples from your potential water sources

Once you have located the water source, the next step is to get samples and have them tested prior to use. Just because your potential water source is called fresh water, it doesn't automatically mean that you can drink it directly without undergoing any filtration or purification process. This goes for all sources, regardless of physical appearance (murky or clear). Subsequent testing at least twice a year - during the spring and fall seasons is highly recommended.

The whole process is straightforward. Begin by securing the approved sampling containers that contain stabilizing agents. You may get them from your local water -testing center or you can secure your own. Again, you can talk to the local water treatment expert for the specifications of the containers.

If you already have a sample container, go to your collection tap. Remove any filtering device. Turn the faucet on and allow the water flow for at least two minutes prior to filling your container. Make sure that your container cap is free from any dirt, grime, or any other impurities. This may interfere with the test results later. As soon as the container is filled, refrigerate your samples right away, and then submit it to the lab within two days.

In the lab, different analyzes are carried out to verify the characteristics of your water source. Here, the presence of pathogens and degree of impurity are established. Determination of total dissolved solids, presence of bacteria, and toxic minerals are all part of the series of tests. According to the United States Environmental Protection Agency, the basic parameters that analysts

use to describe water quality are the following:

- **Temperature:** The temperature of your water source can provide insight into the types of animals that can thrive in such an environment. Extreme temperatures, either freezing cold or boiling hot cause some organisms to die. Temperature also influences the chemistry of your water source. Typically, warm water can hold lesser oxygen compared to cold water.
- **Dissolved oxygen:** This parameter indicates the amount of oxygen available for aquatic species to survive. In general, healthy water bodies have high dissolved oxygen. Factors that affect the levels of dissolved oxygen in the water include: temperature; flow speed; population of algae and other oxygen-consuming species in the water; and, the general profile of the stream bottom (Is it muddy? Is it covered with big rocks or small stones?). The presence of large rocks intensifies the swirling motion of water, allowing enough oxygen to dissolve in it.
- **Nutrients:** Nitrogen and phosphorus are two of the nutrients measured in water quality tests. These compounds are essential for the growth and development of various aquatic species when received in adequate amounts. Excessive nutrients in water result in the proliferation of algal blooms. During the night, algae suck the total oxygen content of water causing the suffocation of fish and other aquatic species. When algae die, oxygen-breathing microorganisms decompose them, again using up the dissolved oxygen in the water. The presence of nutrients affects temperature, pH, and clarity of water.

- pH: pH is a measure of the hydrogen concentration in the water. The p is shorthand for *potential of* while H stands for hydrogen. Typically, pH values range from zero to fourteen, where 0 indicates high acidity while 14 indicates high basicity. A pH of 7 means that the liquid is neutral, not acidic or basic. For water, the typical range is 6.5 to 8.5. Variations in water's pH influence how chemicals dissolve in it and which organisms thrive in the current conditions. Highly acidic and highly basic water are detrimental to aquatic and human life.
- The presence of Bacteria: At some point, you might have heard of fecal coliform, particularly when water-related outbreaks occur in an area. Well, these bacteria are known to reside only in the intestinal tract of humans and animals. They are not harmful in their basic form but in the presence of pathogens and viruses, they can cause health problems. Water contamination due to fecal coliform is caused by the entrainment of animal waste in surface run-off, inefficient septic systems, and discharge from wastewater treatment facilities.
- Toxic substances: Petroleum, metals, and pesticides are some of the most common toxic substances present in contaminated water sources. Lab analysts test for mercury, for instance, in some fish species and in lakes. Mining and power plant emissions are sources of mercury. Without applying the appropriate control measures, these activities can contaminate significant portions of water bodies and the surrounding areas. Once the presence of mercury is confirmed in the lab, people are cautioned not to fish in the contaminated stream.

- **Turbidity:** Turbidity is a measure of water clarity. Lab analysts are able to determine the amount of floating particulates by taking a turbidity reading of your water sample. Imagine yourself in the middle of a pond during the summer months. You might have noticed that you can see its bottom, including the fish that swim freely in it. This is an example of low turbidity. Now, if you visit the pond after a storm, you might not be able to see through to the bottom. Wind movement and the force of raindrops cause soil particles to float around turning the pond into a highly turbid fluid water body.
- **Visual survey:** The physical aspects of the surrounding landscape can affect the quality of inland water bodies. Types of trees and the amount of vegetation covering the ground play a crucial role in trapping pollutants and regulating water temperature. Shaded streams, for instance, provide a cool environment for trout to survive.

The following table from the US EPA shows the current water standards in terms of the parameters above. You can visit the EPA website for a complete listing of all water contaminants, their long-term effects, and possible sources.

Contaminant	MCLG[1] (MG/L)	MCL[2] or TT (MG/L)	Possible Long- Term Health Effects due to above MCL	Potential Contaminant Sources

			exposure	
Total Coliforms (includes E. Coli and fecal coliform.)	zero	5.0% ⁴	indicates the presence of harmful bacteria	Human and animal waste
Turbidity	n/a	TT ³	Measure of water clarity and filtration effectiveness	Soil runoff
Viruses (enteric)	Zero	TT ³	Gastrointestinal diseases	Human and animal waste
Arsenic	0	0.010 as of 01/23/06	Circulatory system problems and increased cancer risk	Electronic production waste, natural mineral deposit runoffs
Asbestos (fiber > 10 µm)	7 million fibers per liter (MFL)	7 MFL	development of benign polyps in the intestine	Asbestos cement decay in water systems
Mercury	0.002	0.002	Kidney damage	Landfill and mining runoff

Source: <http://water.epa.gov/drink/contaminants/>

As soon as the lab completes all your testing requirements, the results are forwarded to you after a few days, depending on the number of tests conducted. Analysts in water testing labs generally have an idea of the common problems present in the various water sources in an area. They should be able to give a proper recommendation to the residents or prospective land buyers.

Primary Water Sources

Based on the water cycle, there are generally three major sources of freshwater: surface water, rainwater, and groundwater.

1. Rainwater

Clouds are composed of tiny water droplets suspended in the air. Upon contact with dust or salt particles in the atmosphere, these water droplets deposit on the surface and accumulate upon collision with other particulates. Over time, the water droplets grow in size and fall at faster speeds, overcoming the updrafts that support clouds. As a result precipitation occurs. Precipitation is the general term that refers to the atmospheric discharge of water either in solid or liquid form. Rainfall is a form of precipitation that results from the condensation of water droplets in the atmosphere, as part of the natural water cycle. It replenishes the freshwater reserves of the earth to sustain plants, animals, and humans.

As a naturally-occurring resource, rainfall does not contain any chlorine, often found in an urban water supply. It is essentially mineral-free, requiring minimal treatment when used for washing, bathing, and cleaning. Therefore, frequent use of rainwater does not result in the scaling of pipelines and appliances.

Despite these qualities, the risk of water contamination remains. Contact of

rainwater with roof materials, piping systems, and storage tanks increases the potential for bacteria and dirt accumulation in the reservoir. Having said that, it is still important to run the collected water through a simple disinfection process to maintain its quality and ensure safe consumption.

The total amount of rainfall that you can collect in a year depends primarily on your location on the globe. While arid areas are generally dry throughout the year, rainfall still occurs, albeit at a lower frequency compared to temperate regions. Cherrapunji in India holds the world record for the highest amount of rainfall in 1861, averaging up to 905 inches within the year. In contrast, Chile, I Africa experienced the absence of rainfall fourteen years in succession.

Because rain occurs naturally, it is a practically free resource. When it rains, it blankets any outdoor surface; you can simply bring out a bucket and leave it there until it fills up. A more efficient way of collecting rainfall is to utilize your roofing system to divert the water towards your storage facility. Having an idea of the total area of your roofing can give you an estimate of the total amount of rainfall that you can collect on a rainy day. Designing your roof such that the rainwater is diverted from the gutter to cisterns and water barrels will significantly increase your recovery.

Please take note of the type of materials and coating that you have used in your roofing. Substances such as lead, which is present in paint, can potentially contaminate your rainwater and make it unsafe for human use. To be safe, be sure to have your water tested at any reputable laboratory to verify the purity of your stored water. Although it is a free resource, some

government units restrict collection or harvesting of rainfall. See if authorities in your area implement this mandate to prevent any misunderstanding or violations.

2. Groundwater

In the USGS depiction of the water cycle shown previously, you will notice that a portion of precipitation infiltrates rocks and subsurface soil.

Essentially, all water sources located beneath various landforms (plains, hills, plateaus, etc.) are generally categorized as groundwater. It can lie a few inches closer to the land surface, such as in marshes, or lie deep up to a hundred feet below the ground. Typically, water in shallow depths is a few hours old. Water at moderate depths may have been there for more than a hundred years while water at extremely great depths may have traveled long enough and settled in the area for more than a thousand years.

For most people, it can be difficult to imagine the existence of water underground. At its most basic form, groundwater is just the subsurface water that accumulates within the void spaces of rocks and soils. It is replenished through the infiltration of rainfall and melted snow into the land surface.

Initially, water that enters the ground replaces the amount absorbed and transpired by plants during the previous dry period. Any excess water trickles down further until it reaches the water table – the top portion of the zone where the voids in between rocks and soil particles are partly filled with air and with water. Beneath the water table, all voids are filled with water that

moves from the aquifer to the well, stream, or spring where you can withdraw it.

The natural process of replenishing aquifers is long and slow because it takes time for the water to move through the openings in the different layers of the soil. More importantly, the amount of precipitation in an area determines the rate of replenishment. Regions that receive lower levels of annual precipitation will have to wait for hundreds of years to refill their aquifers. On the other hand, shallow aquifers in places that receive substantial precipitation can be refilled within shorter periods.

Ground water is generally less vulnerable to contamination compared to surface waters. Layers upon layers of rocks and soil serve as filtering mediums, screening out bacteria and harmful particulates. Despite this fact, you cannot disregard the dissolved minerals and organic substances present in the water although invisible to the naked eye. While some are harmless (such as potassium, calcium, and bicarbonate), the risk of toxic substance contamination remains particularly if the area is near landfills and wastewater treatment facilities.

Water with more than 1,000 milligrams of dissolved minerals per liter is unfit for drinking and is classified as saline. Highly mineralized water is hazardous to human, animal, and plant health. Excess sodium, for instance, is harmful to individuals with sensitive heart conditions. Slightly high concentrations of boron can be detrimental to vegetation. Furthermore, too much calcium or magnesium cause scaling in pipe systems and storage tanks.

Building wells is the one way to access groundwater. In the early days, hand digging of wells is common in households. With a shallow depth for easy water access, wells of the early days are typically three feet wide and installed with a bucket. If you still have this on your property, keep it covered to prevent microorganisms, stray animals, and your pets from contaminating this precious source. It is a perfect water source for emergencies such as drying up of nearby springs. Currently, wells are dug using drilling machinery thus they are distinctly deeper and wider. The typical diameter of modern wells is four to six inches and cased with a PVC pipe.

3. Surface Water

Surface water bodies are perfect water sources both for emergencies and for daily use. Springs, ponds, lakes, and streams are the common examples of surface water sources. The amount of water in these reservoirs varies depending on the inflow and outflow rates. Inflows include precipitation, surface runoff, groundwater seepage, and tributary flows. Outflows (for example, from lakes) include the evaporated water and the discharge into groundwater reserves.

When raindrops hit impervious grounds or drenched soil, the rainwater begins to flow to low-lying areas and deposit into streams until it reaches the ocean. This is called run-off. As it moves over surfaces, run-off carries with it soil particles and sediments that eventually end up in your freshwater reservoir. Most of the water in rivers, streams, and springs come from precipitation run-off either from rainfall or from melting of snow.

Land topography, geology, and meteorological factors influence the amount of surface run-off that goes back to existing freshwater bodies. On the average, a third of the total precipitation amount goes back to rivers while two-thirds evaporate, transpire or infiltrate the soil. Homesteaders often replenish their water supply with surface run-off collection systems in their homes.

In some areas, local government units may restrict collection of surface run-off or even utilization of inland freshwater sources. You can check with your local units to verify if it applies to your area.

Harvesting Water

There are several options when it comes to providing a safe and plentiful source of water for your homestead. Digging a well and collecting rainwater are probably the two standard methods of water collection that are frequently implemented in off the grid living scenarios. It is also a good idea to become familiar with some water collection methods that have been devised by survivalists for use during emergencies.

1. Rain Collection

Collecting rain is an ancient method of gathering water. Devices for collecting rainwater that are over two thousand years old have been found in various places around the world. Evidence of rain catchment systems in roof designs were discovered in ancient Roman villas and cities, proving that rainwater was a vital resource for drinking and household use.

For areas receiving little annual precipitation (average of 100 mm) such as the Negev desert in Israel, storing hillside runoff after the rain allowed them to thrive in this dry region. They had sufficient water for domestic use; for raising livestock and maintaining the agricultural land. In Egypt, rainwater storage tanks with capacities ranging from two hundred to a thousand cubic meters, built more than a thousand years ago are still functional to date. The largest rainwater storage is located in Yerebatan Sayarin in Turkey. Built during the reign of Ceasar Justinian, it is supported by more than 300 columns, each having a towering height of 30 feet. It has a capacity of 80,000

m³, measuring 140 meters by 70 meters.

Harvesting, storing, and utilizing rainwater has several advantages:

- it promotes the conservation of groundwater sources;
- it delays the need for expensive water utility expansion projects by minimizing water demand particularly during the summer season;
- it maximizes the use of storm water runoff;
- it consumes lesser energy in comparison to wells;
- it provides a viable water source for various off-grid households

With the right design, use of standard construction procedures and regular maintenance, a rainwater system can adequately supply high-quality water throughout your homestead life.

Types of Rainwater Harvesting Schemes

There are three fundamental elements in a functional rainwater harvesting system: a) collection system, b) transporting system, and c) storage system. A collection system answers the question: How are you going to gather the rainwater? It could be as simple as leaving empty buckets outdoors until they

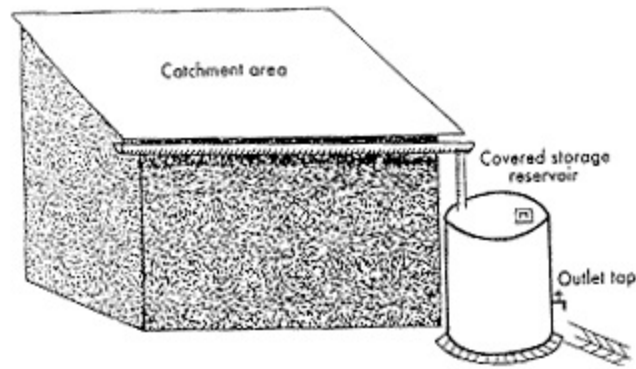
fill up or you can use your roofing surface and gutters as temporary catchment areas. When thinking about collection systems, consider your water requirement and your locality's precipitation pattern to come up with the most effective mechanism for your homestead.

Unless your catchment device acts as your storage equipment as well, you need to find a way to divert the water to a covered containment device. Downspouts are often used to divert water to cisterns and tanks. A conveyance system can be simple or complex depending on the capacity and design of your rainwater harvesting system.

Finally, storage systems contain water for long durations until it is used. These are installed either above or below ground. More information on the types of storage will be discussed in the next chapter.

a. Roof collection system

The best way to collect rainwater is to install gutters on the roof of your house or other building and funnel the water into barrels, where it can be stored until needed. There are commercially available gutter systems designed especially for facilitating the collection and storage of rainwater. Several barrel styles are available for purchase. It is also possible to construct your own rain barrels.



Source: <http://www.unep.or.jp/ietc/publications/urban/urbanenv-2/3.asp>

If you are looking for a simple, no-frills way to gather rainwater you can do so easily with a tarp and a container system. Set up the tarp and poke a hole in the middle. Place the container underneath the hole. You may consider a length of pipe to help the water flow into the container. When it rains, the tarp will act as a funnel and the rainwater will pour into your barrel.

Collected rainwater is never clean enough to drink or use right away. It must be filtered to remove insects, bits of plants, and other debris. Again, make sure you test the water and make the necessary purifications prior to using or drinking it.

The Three Classifications of Roof Collection Systems

According to the Rainwater Harvesting Best Practices Guidebook, published by the Regional District of Nanaimo in British Columbia Canada, roof collection systems are classified into three main categories based on the usage of the water:

- a. Non-Potable Water for Outdoor Use: includes irrigation of your crops and garden plants; filling the pool; water for livestock; outdoor cleaning; washing of vehicles; and fire protection
- b. Non-Potable Water for Indoor Use: includes toilet flushing; indoor cleaning
- c. Potable Water: includes water for drinking, cooking, bathing, washing the dishes, clothes washing.

Each category has its own set of components, operational conditions, and maintenance requirements based on the water quality required for their respective purpose.

Non-Potable Water for Outdoor Use

This type of rainwater collection system operates seasonally and is often shut down during late fall. It requires less stringent filtration techniques for the removal of sediments and bacteria. Maintenance involves occasional inspection and cleaning of pipelines leading to the storage facility.

In general, any standard roofing, gutter, and downspout work well for this rainwater collection system. When designing your roof and catchment systems, be on the lookout for items that may contain lead and other metallic contaminants. Install debris traps along the pipeline to remove entrained leaves or pebbles. This practice minimizes the tank cleaning frequency and prevents the proliferation of algal blooms particularly if the tank is directly exposed to sunlight.

A water diverter system is a smart way to keep the water quality in your storage free from contaminants, especially when it is time to clean the gutters and pipelines. It can be as simple as closing a valve so that water immediately flows out of the diverter pipe instead of the tank pipe and drains away from any underground reservoir.

Non-Potable Water for Indoor Use

Previously, you have seen that toilet flushing consumes 18% of the total water supply in a typical American home. In British Columbia, the amount is higher, accounting for almost 30% of the total water consumption. Thus, dedicating a separate system for this activity is a sensible approach.

Maintaining and running non-potable rainwater systems for indoor use requires more work and keener attention to detail compared to the outdoor systems. Because you need water all year round for flushing, you need to check for any signs of debris buildup frequently. Debris removal minimizes water discoloration. Installing filters at the tank inlet and discharge points will also keep sediments from contaminating the water.

Standard roofing materials, downspouts, and gutters also work well for this type of rainwater collection system. Avoid cedar shingles due to their inherent toxicity and capability to bleed color into the water.

Pay attention to any organic material build up and remove any debris as soon as possible. Additional filters before and after the catchment system can significantly reduce suspended debris before the water enters the tank.

Non-potable water systems for outdoor and indoor water use typically have small volume tanks ranging from a thousand gallons up to five thousand gallons. It can be installed either below or above the ground and must have an overflow pipe for draining any excess water.

To prevent mixing of your potable and non-potable water reserves, install separate distribution lines for each system. You can use colored pipes for easy identification. Provide proper labels for their respective taps to eliminate any confusion and minimize health risks.

Potable Water

Because potable water needs to be safe for human ingestion, you must adopt more stringent measures when collecting rainwater. Potable systems are run all year round to provide sufficient water for drinking, bathing, cooking, dishwashing and doing laundry. For this reason, potable water storage facilities have higher volumes ranging from 8,000 to 15,000 gallons, compared to non-potable water storage facilities.

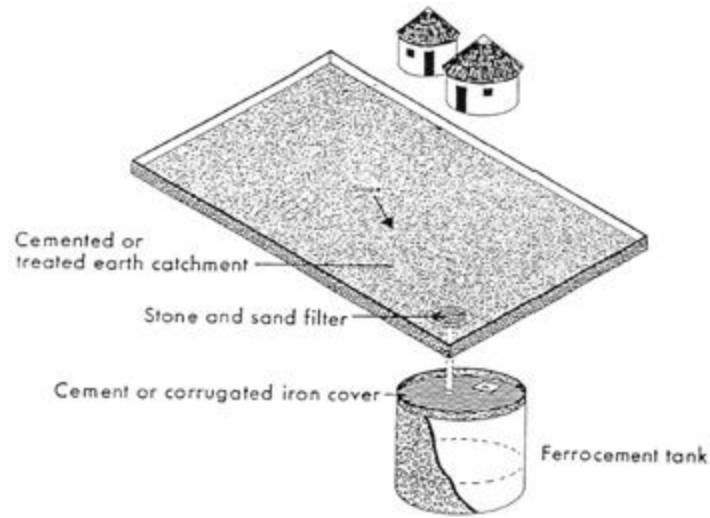
Tile rooftops are your best bets. While standard downspouts and gutters still work well for potable water systems, you need to choose carefully designed parts that minimize water and debris contact. Use gutter covers to prevent entrainment of organic materials and other potential contaminants. Also, look for tanks, pipes, and pumps that are suitable for potable water storage and transport.

Compared to non-potable water systems, more filters are required to remove any suspended debris in the water from catchment to storage until end-use. Available filtration devices include fine screens, underground water filters, and gutter covers.

In addition to debris removal equipment, use of disinfection devices inactivates viruses, bacteria, and other harmful microorganisms present in the water. If you are unsure what to buy, ask the experts and choose only certified equipment. While setting up your own potable water system is expensive, the rainwater, on the other hand, is practically free! In time, you will be reaping the unending benefits of having natural access to safe and pure water in your homestead.

b. Ground surface catchment

If you have a large land area, you can use a portion of your property to collect rainwater. Compared to rooftop catchment systems, ground surface catchment systems have more surface area allowing collection of greater volumes of rainwater. Aside from rainfall, you can also retain surface runoff such as streams and flood flow; augmenting your water supply even during the dry months.



Source: <http://www.unep.or.jp/ietc/publications/urban/urbanenv-2/3.asp>

Water loss due to ground infiltration is one disadvantage of this method. You will also need a more efficient filtration system since the risk of entraining contaminants is higher in this case. You will need to apply more stringent measures if you intend to use it for drinking. However, using simple filtration methods, such as the use of sand and stone filter, are sufficient if you plan to use the collected water for agricultural irrigation or cleaning purposes.

How to Build Your Own Rainwater Collection System

This simple system makes use of a 275-gallon container popularly called the IBC for water storage. Popular rain barrels can only store up to 55 gallons of water. With the IBC, you can collect five times the typical rain barrel volume while consuming only a portion of your backyard space.

For this project, you will need the following:

- A used IBC that you can buy from local syrup bottlers or from Craigslist
- The IBC cap, 6-inches in diameter
- a 4-inch cap
- A used pallet of a few pieces of wood for the platform
- A gutter screen
- Downspout, elbow, and gutter
- Silicon sealant
- Fiberglass bug screen

The first step is finding a location for your storage tank. It has to be near the downspout to minimize unnecessary pipelines from catchment to storage, and must be placed at a lower elevation relative to the catchment for ease of water flow. Once you have chosen a good spot, you can place a used pallet (if the wood is still in good condition; without rotting portions or signs of termite attacks) on it to slightly elevate the tank and keep the IBC tap from touching

the ground.

The second step is to modify the IBC inlet such that any form of debris is left outside as the water enters that tank. For this part, take out the six-inch screw cap of the IBC and center a 4-inch diameter pipe (around 2 inches thick) or pipe cap. Trace the outline of the 4-inch pipe and use a hole to create an opening. Next, drill holes around the inside edges of the caps and use rivets to keep them together.

In the third step, use a 3-inch hole saw to create an opening in the 4-inch cap. Get the gutter screen and trace the outline of the 4-inch cap onto it. Using the tin snip, cut out the circle using the inside of the line as your guide. Fit the screen inside the 4-inch cap. You can use sealant to secure the screen in its place.

For the fourth step, cut out a portion of the fiberglass screen. It should be a bit larger than the cap. Lay it over the 6-inch tank inlet then screw the modified cap on it.

Now, that the inlet is done, it's time to modify the outlet. The IBC tank is equipped with a 2-inch drain port with a plastic shut-off valve and a 2-inch screw cap to keep it sealed. For the fifth step, simply drill a $\frac{3}{4}$ -inch hole through the middle of the cap and install a $\frac{3}{4}$ -inch faucet through it. The NPT male type is preferred over the standard threaded type because it maintains a uniform diameter as you screw it further. This ensures a watertight connection. Seal the connection with a waterproof sealant and tighten a nut

around the back portion of the faucet. Now, connect a hose and test the flow of water.

The sixth step is about modifying your existing downspout such that it is directly in contact with the inlet of the tank. If your tank is placed right next to the downspout, you can cut a portion of it so that its outlet is only a few feet from the tank inlet. Use the elbow and the cut-off part to connect the downspout end to the tank inlet.

And you're done! Clean the pipes, caps and gutter thoroughly. As you do so, observe how water flows and see what you can improve further (example: sealing, angle of water drop-off point, etc.). When the rainy season comes, you will be ready to harvest the water in no time!

Possible Legal Issues with Rainwater Collection

You would think that gathering water that falls from the sky and using it to facilitate your survival would be an activity that is possible to engage in without the interference from local and state governmental agencies. Sadly, this is not the case. More and more areas are passing or trying to pass laws that either force a tax on collected rainwater or make it illegal to do so in the first place.

The fact that a government body thinks it owns the rain and can fine or tax you for using collected rainwater is almost reason enough to get off of and stay far away from the grid. Local governments have a track record of

hassling those who choose to turn their backs on the grid and seem to be trying to make it as illegal as possible to live a natural life. It's good policy to research your local ordinances before collecting rainwater to avoid legal entanglements.

2. The Well

Digging or drilling a well on your property is hands down the best option for ensuring a safe and consistent source of water that can meet the needs of your property and lifestyle. Under normal conditions, a well can provide upwards of two gallons of water per minute. The water is usually safe to drink and consume without undergoing a treatment process.

Wells are generally classified into one of two categories, depending on the kind of water they provide. "Well Water" comes from a well that is often over two hundred feet deep. These types of wells are usually professionally installed and can run upwards of three or four thousand dollars to dig and construct. This cost is minuscule compared to a lifetime of water bills, and this option should be considered as a first choice for anyone seriously considering establishing a proper homestead. The water from this type of well has the added advantage of almost always being safe to use as soon it comes out of the well. In only the rarest of conditions will you have to be concerned with purifying and treating the water. Of course, it should be tested prior to use to make sure it is pure enough for direct consumption.

Types of Wells

Wells that provide what is called "groundwater" are most often under fifty feet deep. Because the water supplied by a shallow well is more prone to contamination, it is often necessary to treat or purify the water in some way.

A shallow well works great as a way to augment your other water collection efforts, give serious consideration before choosing this type of well as a primary source of water. In some cases, it is best to suck up the extra cost and have a professionally drilled well installed on your property.

Anyone who has the time and inclination can dig their very own shallow well. Because it is possible to install this type of well on your own (or with the help of family and friends), and the well isn't going to need to be that deep, the cost of constructing and maintaining a shallow well will be considerably less than a well that provides well water. The downside to these savings is the questionable quality of the water that you are pulling from the well.

Another way to classify wells is according to the type of construction. The North Carolina Cooperative Extension Service describes five methods commonly applied in their locality:

- **Dug wells** have large diameters typically around 24 to 36 inches and are excavated with picks and shovels. You will often find these wells lined with a mixture of bricks, clay, rocks and concrete starting from the ground to a depth of 30 feet. The period before the Second World

War was the heyday of dug wells.

- **Bored wells** are the modern versions of dug wells. Instead of using picks and shovels, people use augers for quicker excavation, allowing the bored well to reach up to a hundred feet in depth. While it is a more efficient method of digging, it causes minimal disruption in the immediate landscape. A concrete pipe or well tiles provide a solid, sturdy casing for keeping dirt, underground creatures and other sources of contamination out of the well.
- **Driven well** construction entails driving a steel pipe, usually with a small diameter, installed with a screen and drive point, into the ground. A shallow driven well can be installed by any homeowner or farmer in his own piece of land provided that the hydrogeological conditions are favorable (abundant water at shallow depths, sandy soil, or clay soil with minimal rock layers). Compared to professionally driven wells, DIY-driven wells may lack sufficient protection, particularly if the builder does not have the right equipment, experience, and expertise to do it. Before driving your equipment into the earth, conduct thorough research and consult with professionals to know the basic guidelines for constructing a sturdy driven well.
- **Jetted wells** are uniquely constructed because the method entails utilization of water pressure to dislodge the soil in an excavation. If the chosen well site contains dense materials (sandstone or slate) installing a chisel bit can aid the jet in breaking up the solid particles.

- **Drilled wells** are excavated through rotary or percussion methods. In percussion drilling, a heavy chisel bit is repeatedly dropped until the existing rocks and soil break apart. Rotary drilling, on the other hand, relies on a rotating drill bit to penetrate through the rocks and soil. The principle is similar to hand-held drill bits.

In building a well, there are three factors to consider: the flow rate, static water level, and last but not the least, the depth.

Depth pertains to how deep the drillers must dig to reach an aquifer. Flow rate pertains to the volume of water that seeps to the well per minute. The static water level refers to the highest level the water can reach after breaking through the barriers on top of the water table.

Consideration of all these factors will ultimately determine the cost and the feasibility of your well-building project. Before proceeding with building your own well, you need to have an idea of the daily water demand in your household. After that, you can proceed with locating aquifers within your locality and conduct an initial assessment of the amount of water in it. Of course, you also have to consider how fast or slow it is replenished to verify its sustainability in the long run.

Locating the best place to dig a well is the most important step in the process. It is worth the time and effort to have your land professionally surveyed to assist you in choosing the most advantageous spot. A geological survey gives

you a big picture of the underground features of your current property. You can get a lot of information from the results such as the type of rocks beneath the surface, composition of the soil layers, and the availability and location of potential groundwater sources. It is important that the well is located away from sewer lines, septic tanks, and other possible sources of contamination. Building a well fifty feet away from the sewer and service lines is one of the best practices.

How to Build Your Own Drilled Well

While drilling rigs are more often used by professional well drillers, current technology can help you drill your backyard well. Toolkits that utilize air driven motors to push the drill bit into the ground are commonly available and effective. One set contains a well point, a drill bit, air-powered water pump, and some instructional materials to help you get started.

Let's say you've finally found the perfect spot for your well. Before proceeding with the use of the toolkit you need to do some preliminary well prep. Start by digging a hole either with a post digger or with a hand-held auger. Make sure that the hole is vertical and dug as deeply as possible to save on some driving distance. A safe indicator is until you have reached the point where water comes out to keep the hole filled constantly. The water serves a dual purpose. First, it acts as the cooling agent for the drill and motor to prevent overheating due to friction between the ground and drill. Second, the presence of water makes the mud less viscous and easier to pump out of the hole.

To make the air-powered drill work, you will need an air compressor that can produce at least sixteen cubic feet per minute (maximum of 25 cubic feet per minute) of air at 90 psi. Drill up to the target depth (at least 100 feet to 300 feet). After drilling, install a PVC pipe, about four inches in diameter. This will serve as the casing of the well. Take note that the first segment of the pipe needs to be perforated so it can function as your well screen. Place the well point at the end of the perforated segment to plug the pipe and guide it down the hole. It will also protect the casing from crashing into the bottom of the well. Once you have lowered the casing, fill the void spaces around it with pea gravel. Use a cement cap to prevent any debris from entering the well hole.

Now, to pull the water upwards, you need to install an underground pump. There are types that suitably fit inside a four-inch diameter pipe and are typically powered by electricity. Solar and wind-powered pumps are popular alternatives among homesteaders.

After building the well, the first hundred gallons of water that come out are most probably muddy. Don't worry, this is normal. Eventually, the water will become clear as the soil particles settle. Just make sure to have the water tested before using it.

3. Springs and Other Natural Surface Water Sources

Springs

Springs are points where groundwater converges and discharges to the surface. It can form on hillsides, valley bottoms, or other spots where excavations intersect the flowing water exactly at the water table or below it. Many homeowners are able to locate springs within their properties. One dominant characteristic of a spring is that it naturally surges from the ground. You can also find your very own spring with the help of these tips:

a. Locate your spring during the dry months

Choosing dry months to locate your spring will help you locate it sooner. When the earth is saturated with rainwater, it is harder, if not almost impossible, to track watermarks within a potential spot. Consider the summer season as your best option if you want to increase your chances of finding a spring.

b. Seek for an area with middle elevation

Areas with middle elevation are the perfect spots for finding a spring. An example is the side of a hill. This is where the head of the spring can be found. Since it has a bit of elevation, it carries with it potential energy. When you dig into this area, gravity pulls on the water and forces it to flow down from its location. You can then install a simple piping system to divert the water to your filtration system before utilizing it at home.

c. Keep an eye out for signs of erosions

Signs of erosion can indicate the presence of a spring. Since the water from the spring flows down from its elevated location, it naturally carries with it some soil particles causing the erosion within the vicinity of the property. Moss growth and smooth rocks are also great indicators of an abundant water flow in a particular area, where the spring may flow.

Tapping into a spring is easy. After finding a head, the next step is to install a piping system. Since the head is typically in a slightly elevated area, you can use the force of gravity to bring the water down to your residence. In some cases, the water pressure may not be enough to provide flow. You can install a water pump to address this problem. Various sizes of pumps are available in the market. Ensure that your chosen pump is designed for your system to avoid excess costs, present in buying an oversized pump. Solar pumps are also a great option to save on electricity costs. Just ensure that the area where the pump will be installed receives sufficient sunlight throughout the day to optimize its use.

Creating a water reservoir allows the water to build up before it goes to your home. This will ensure continuity in water flow compared to directly diverting the spring water to the house. Since the flow rate of the spring may fluctuate, depending on the availability of the water, you might experience shortages from time to time, particularly on dry days. If you find a small spring, you can create a small reservoir around it. Allow some time for the reservoir to fill before diverting it to your storage.

Upon turning your faucet, you might notice that the water is initially murky.

Allow some time for the soil particles in the spring water to settle – typically for a few days if you don't have a filtering system, before drinking. For bathing and toilet flushing purposes, this water is generally safe.

Your spring water source may also need a bit of maintenance each year. Sand and gravel build-up in the piping system and sinks are commonly encountered by homeowners through the years. One strategy to avoid this problem is to install adequate screening systems and filtering devices that will remove these impurities in the water before it enters your home. Regular checking of the screens for holes and other damage is an excellent preventive measure that ensures the purity of your spring water for a long time. Inspecting the head at least once each year will make you aware of any pumping problems and leaking issues in the piping system.

Ponds

Wells and springs are convenient natural sources of water, provided that your location allows you to have ease of access to these resources. In cases where groundwater is not easily obtained, ponds are great alternatives. In fact, ponds are gaining popularity in relatively dry areas, such as North Idaho. Water from their ponds supplements their farm irrigation and fire control requirements. In this section, the use of ponds as a storage structure is given focus.

Water in ponds is sourced from various bodies of water. While ponds are a great way to store water for general use, drinking from it may cause harm to

human health. Usage for livestock rearing, vegetable farming and fish farming are some of the common uses of pond water. Since livestock manure is mixed with the water, as in the case of fish farming, possible breeding of bacteria and other pathogens may render the water unsafe for human ingestion.

Typically, ponds have a miniature dirt dam that intercepts the water flowing from higher elevated areas after raining and snowing. To keep the water level of the pond high, there must be sufficient flow of water from the uplands. Thus, locating a suitable area for a pond entails knowing the precipitation density in a particular area including the elevation and slope of the land in consideration. You can also think of the uses of the water in your ponds. Do you need it for gardening, livestock rearing, or household use? Establishing your purpose will also help you decide on the most suitable location for your pond.

Constructing a pond requires removal of loads of dirt. Using appropriate machinery speeds up this process. You can either consider delegating the work to a third party or you can purchase your very own equipment. Both choices have their own set of advantages and disadvantages. Contracting someone with the digging machinery may be costly but you won't have to worry about the nitty-gritty of the work. If you prefer DIY projects, buying your own equipment and doing the digging yourself may delight you. An important consideration is if there will be other uses for the equipment. Otherwise, it may be impractical to purchase the equipment even if you go with second-hand, slightly used machinery.

Creeks

Any flowing water body bounded by banks is generally known as a stream. In this case, rivers and creeks are variations of streams. Compared to rivers, creeks are smaller forms of streams that empty into another reservoir or flowing water body.

Surface runoff replenishes creeks. Some creeks have shorter life spans and dry quickly. This is common for shallow creeks. Others retain a consistent amount of water throughout the year. It is common to find freshwater creeks although salt-water creeks also exist.

You may have some luck digging in a dry creek bed for water. Usually, a relatively deep hole is needed to get past the muddy, murky area of the creek bed to reach a source of acceptable water.

Springs, ponds, and creeks are all potential sources of drinking water. Even if they are known as freshwater sources, it should be assumed that they are contaminated and unsafe. Proper treatment is always required before using water from these natural sources.

4. Dew Collection

Collecting dew won't ever provide enough water to run an entire household or homestead but it will provide you enough to get by and stay alive if you need to. The method is rather simple. You just need to soak up the dew from

plants with a cloth in the morning. Then wring the cloth out into a container.

The above-mentioned methods of water collection will if properly implemented, provide you with all the water you need to effectively run your home. Collecting water is only part of the equation, it still needs to be treated or purified and then safely stored.

Digging a well and collecting rainwater are two of the best ways to make sure you can meet your demand for the delicate mixture of hydrogen and oxygen that we call water. There are a few other methods of water collection that are worth knowing. While they are not as reliable as a well, they are useful methods to know about in the event of an emergency or should your primary water source be temporarily out of commission.

Chapter Three

Water Treatment

Once you have decided on how you are going to provide your homestead with water, you'll need a way to ensure it is clean enough to safely use. Water can be purified in several ways. Common methods include using chemicals such as bleach or iodine to kill infectious agents, various filtration devices to remove impurities (including viruses, bacteria, protozoa, etc.), and boiling or otherwise heating the water to destroy dangerous contaminants.

Most of the time, a mixture of chemical treatment, filtration, and applying heat is necessary to provide you with water that is fit to be consumed. Filtering water is often the first step and may even be done a few times before everything is said and done. Rainwater is notorious for being filled with plant and animal debris, and filtration devices are usually built into the gutters and containers to cut down on the amount of twigs and dead bugs you find. Electric filtration devices are available, and it would be a good idea to think about purchasing one. Water purification is not a good area to cut budgetary corners on. Hand-held water filtration devices are also available, but they are not capable of handling high volumes of water and are unfit for meeting the needs of an entire household and/or small farming operation.

Making a Water Filter

You can also make your own water filter for when you are in an emergency situation or as a fun science project for you and any kids. Constructing your own water filter is a good way to see how the whole process works.

Materials

To make a water filter you will need:

- **Two Containers** - Five-gallon water bottles or a similar container is an excellent choice. One of these containers will hold the filtering agents and the unfiltered water, the other is to capture the filtered water as it drips from the filter container.
- **Rocks** - These will form the base of your filter and will be placed at the bottom of your filtration container. They will need to be heated in a fire first.
- **Sand** - This is the next layer of filtering material you'll be adding this on top of the rocks.
- **Charcoal** - You can use commercially available charcoal or burn some wood in a fire until it is black and charred, then smash it into a fine

powder. Both are acceptable methods of acquiring charcoal.

- **A Fire-** You'll need this to heat your works and possibly burn some wood for charcoal.
- **Silver** - Either fine dust like particles or old silver coins will do. In the United States, coins minted prior to 1964 are made mostly of silver and will work just fine for this project. Silver has strange chemical properties that make it effective at destroying microorganisms and harmful bacteria. You'll be adding the silver dust (or old coins) to the top few layers of charcoal.



Steps in Constructing a Water Filter

1. Take one of the containers and poke a series of tiny holes in the bottom.
2. Heat your rocks until they are glowing and allow them to cool. Once they cool add them to the bottom of your container.
3. Add a layer of sand, pack it down as best you can.
4. Begin to add layers of charcoal with the goal of filling the container. Be sure to add some silver coins or dust to the layers of charcoal near the container's top.
5. Once constructed, pour some unfiltered water into the filter and allow it to drip down and out the bottom into the other container. Remember, never use your filtered water container for unfiltered water.

Filtering water is an effective method for removing bacteria and protozoa, but many viruses are small enough to slip through many common filter types. Using a filter to remove debris and cut down on microscopic organisms is a practical first step and a good start, but you'll almost certainly want to use another method in conjunction with it.

Using chemicals to treat and purify water is standard practice. Nearly

everyone is familiar with the use of chlorine to keep a swimming pool's water clean and clear.

Chlorine is nothing more than common household liquid bleach, and can, in small amounts, be used to purify water and make it suitable for drinking.

Iodine, familiar to many as a topical antiseptic for minor cuts, is also an effective way to purify water.

Methods of Using Bleach to Purify Drinking Water

Chlorine is known as a powerful disinfectant. It is used to eliminate unpleasant odors, improve water taste and inhibit microbial growth. One disadvantage of chlorine use is the formation of toxic products that are detrimental to human health. To treat water with chlorine, follow these steps:

1. Make sure the water is at least sixty-five degrees Fahrenheit.
2. Add one-half teaspoon bleach to every five gallons of water. If the water is exceptionally cloudy, add a teaspoon to every five gallons.
3. Cover the water and leave it alone. In a few hours, check on it.
4. If the water is clear and the smell has a faint hint of bleach, the water is suitable for consumption. If it isn't, repeat steps two and three once more. If the water is still cloudy or doesn't smell faintly of bleach, then you'll have to give up on it and find another source of water.

Alternatively, you can use commercially available sodium hypochlorite for water treatment.

1. Simply fill a wide-mouthed container with twenty to twenty-five liters

of water. If the water appears to be a bit cloudy, pre-filter the water with cheesecloth for a few times until it becomes clear.

2. Add in the sodium hypochlorite. The amount depends on the specifications of the manufacturer. Typically, if the water is clear, you need to add only one tablet of chlorine. But, if the water appears murky, you need to add two tablets after pre-filtering it.
3. Seal the container with a lid and leave it for half an hour.
4. You can now drink the water.

Take note that the antibacterial effects of chlorine depends on the pH of the water. If it is possible to measure the pH, please do so before proceeding with the treatment. In general, it is important to maintain the water pH between 6 and 7. If the pH hits 7.5 or more, the active hypochlorous acid chlorine decreases in concentration rendering the treatment ineffective. If the pH drops below 6, chlorine gas may evolve and cause irritation in the respiratory system.

Purifying Water with Iodine

1. Make sure the water is at least sixty-five degrees Fahrenheit.
2. Add eight drops of iodine for every half gallon of water. If the water is exceptionally cloudy, add twice that amount.
3. Cover the water and leave it alone. In a few hours check on it.
4. If the water is clear, you're in luck! Your water is now ready for use. If it isn't, dispose of it. It's not healthy to add any more iodine.

Potassium Alum for Water Purification

The use of potassium alum is an ancient practice. It works by reacting with suspended particulates in the water until it coagulates into larger particles that easily settle. A community in Myanmar evaluated the effectiveness of alum by placing a solution with a concentration of 500 milligrams per liter into 160-Liter vessels. The results were astounding. Fecal coliform was reduced by almost 98%, eliciting a positive response from the community.

Despite its effectiveness, use of alum potash is difficult to implement without the right training, according to the World Health Organization.

Using Natural Products

Some communities in the United States use fruit seeds, beans, or cassava to separate suspended sediments from water. The steps are simple:

1. Dry the fruit seeds under the sun.
2. Grind the seeds until it becomes powder-like.
3. Mix 0.5 grams of the powdered seeds with a liter of water.
4. Stir the solution for a minute then leave for two hours.
5. Check if the particles have settled. You can now use the water in the top portion of the container.

A Solar Still

A solar still is a device that mimics the rain cycle. It's a container that is filled with water and allowed to sit in the sun. The sunlight evaporates the water and forces it to condense on a glass or plastic surface. The evaporated water condenses on this surface and becomes purified in the process. Solar stills are commonly used for feed water.

Solar Disinfection of Drinking Water

The sun is known to emit ultraviolet rays can be harmful in excessive amounts. But, did you know that these rays are useful for treating water too? Often used in the African regions, this treatment method requires the use of clear PET bottles (around 3 to 5 liters) as the water container.

1. Wash the PET bottle thoroughly with soap and clean water especially if this is the first time you'll be using it.
2. Check if the water to be treated is turbid. If it is, pre-filter the water using cheesecloth until it becomes clear in appearance. Otherwise, proceed to the next step.
3. Fill the PET bottle with water then leave it exposed under the sun for six hours. If the weather is cloudy, extend the exposure of the water to 2 days.
4. Collect the solar disinfected water. You can drink it or store up to 2 days in a dark area. Keeping the disinfected water in sealed containers reduces the risk of recontamination.

Solar Pasteurization of Water

If you have a solar cooker, you can use it to pasteurize the water. Take note that the pasteurization process will kill all forms of germs, harmful bacteria, or disease-carrying microorganisms but it does not eliminate chemical contamination from pesticides and industrial waste nor does it sterilize the water.

Using Heat

The final method of water purification is to simply boil it. All you need is a pot and any kind of fuel. There aren't too many contaminants that can withstand being boiled for longer than ten minutes or so.

Unlike in chlorination where murky water needs to be pre-filtered before treatment, boiling works well for all sorts of water (except of course heavily polluted water). After filling the pot, allow the water to boil for several minutes. Take note that the duration of boiling depends on the altitude. If you live in a highly elevated area, it will take a longer time for water to boil. At sea level, the water boils quicker due to the higher pressure exerted by the atmosphere. As a guideline, add one more minute to your normal boiling duration for every thousand feet above sea level.

Once you are familiar with the best way to clean and purify your water, the final step is properly storing it. It's a simple process and thankfully much easier than digging a well or treating your water.

Chapter Four

Storing Your Water



Every homestead has its own annual water requirement. It depends on the number of people and the types of water-consuming activities conducted on a daily basis, for instance, the size of your livestock or cropland influences how much water will be used for outdoor activities. Having an idea of how much water you need can help you verify if your current sources sufficiently meet your demand or not. Furthermore, the annual requirement will help you determine what the appropriate size of your water storage system is.

In chapter two, you have learned a simple formula that you can use to calculate your annual water requirement. Chapter 3 discussed all types of water sources that you can tap in your area depending on availability. Here, storage systems will be discussed, including how to size your tank.

Types of Water Containers

Properly storing your collected and treated water is a simple process. It entails nothing more than procuring suitable containers, sanitizing them and making sure they are in a safe place. If a container can be sanitized and hold water without leaking or letting in debris, it can probably be safely used to store water.

Glass jars and jugs make good containers for small amounts of water. Plastic containers made of from materials deemed as "Food Grade" can also be used to store water. Whether you need a small tank or a large tank, you can always find the most suitable container for long-term storage of your non-potable indoor water, non-potable outdoor water, and potable water.

Barrels and steel drums are useful for storing large amounts of water in one container. Usually, these types of storage devices are capable of holding in excess of fifty gallons of water. You can also use barrels as water reservoirs for water heaters.

If you need to keep a large amount of drinking water on hand, consider a water tank. This is an investment in time, money, and requires a permanent structure on your land but is a viable option for many situations.



There are a myriad of containers available in the market to date. Each has its own sets of advantages and disadvantages. So how do you choose the most suitable tank? First, you need to know your storage requirement.

Calculating Your Storage Requirement

For rainwater, the maximum amount you can collect will depend on three basic factors:

1. The average precipitation in your locality;
2. The total roof catchment area; and,
3. The collection efficiency

The precipitation every year varies up to 25% either above or below the average value. Getting at least ten years of data for rain and snow from your local climate monitoring organization, will give you a picture of the rainfall pattern and the estimated amount that you can collect in a particular month.

The total surface area of the roof is different from the catchment area. Unlike the surface area, the slope of your roof does not influence the catchment area. To determine your current catchment area, all you need to do is to measure the following:

- Width is just the distance from one gutter to the other
- Length is just the total length of the gutter

Total catchment area equals width multiplied by the length

For a 9-meter wide roof that is 11 meters long, the total catchment area is:

$$9 \text{ times } 11 = 99\text{m}^2$$

The percentage of rainfall that lands on the roof and finds its way to your storage is your collection efficiency. A good rule of thumb to follow is that you can collect around 75 to 80% of the total annual precipitation. The reasons for water loss are accounted for by the following:

- Site-specific water loss due to the type of roofing material, prevailing wind speeds, and presence of overhanging vines and branches;
- Collection system loss due to overflow and spillage caused by inefficient filters and undersized pipes; and,
- Seasonal factors such as the extension of dry periods, snowfall that remains as ice, and the need to divert water out of the collection system due to scheduled cleaning.

If this is your **first time** setting up a rainwater harvesting system, you can use the annual potential rainwater collection figure to estimate your storage need. To do that, simply use the following formula:

$$\text{Potential Water Collection (liters)} = \text{Total Roof Catchment Area (m}^2\text{)} \\ \text{multiplied by Annual Precipitation (mm)}$$

Take note that 1 mm of rain on a 1 m² catchment area produces 1 liter of water.

Consider this example. The 12-year average annual precipitation at your local weather station is 1,000 mm. Your house has a catchment area of 90 m². This could capture:

$$90 \text{ m}^2 \times 1,000 \text{ mm} = 90,000 \text{ liters (90 m}^3\text{) of water}$$

If you already have existing rainwater storage and would like to know if you need an upgrade, knowing the actual amount of rain that you can collect in a year will give you a heads up on the size of your rainwater storage.

Calculating this amount is simple using the following formula:

$$\text{Actual Water Collection} = \text{Potential Water Collection multiplied by} \\ \text{Collection Efficiency}$$

Going back to the previous example, if your potential water collection is 90m³, and using the rule of thumb, 75% collection efficiency, your actual water collection is:

$$\text{Actual Water Collection} = 90 \text{ m}^3 \text{ multiplied by } 0.75[3] = 67.5 \text{ m}^3$$

If you have or are planning to have an open storage facility like **ponds**, you will need precipitation data for proper sizing. Your best option is to seek help from your local Natural Resources Conservation Service conservationist to find out how much runoff you can collect given the average precipitation data. From this information, you can estimate the dimensions of your pond. Also, check out the water regulations implemented in your area before proceeding with the construction and use of the pond.

Storage Tank Sizing with Water Balance Tables

A water table is a useful tool for homesteaders. It compares the water supply against the total demand to verify the actual volume of water that remains in the storage at the end of each month. The table below is a sample water table for outdoor use; it may differ from yours depending on your current household demand and available supply:

Month	Outdoor Usage, Liters	[Assumed precipitation][4] ,	Assumed Collection Efficiency, %	Collected Rainwater, Liters	Month-end Water Volume
		Millimeters			
Nov	0	154.9	0.8	11,153	11,153
Dec	0	137.1	0.8	9,871	21,024
Jan	0	152.4	0.8	10,973	31,997
Feb	0	109.2	0.8	7,862	39,859
Mar	0	99.1	0.8	7,135	46,994
Apr	0	53.3	0.6	2,878	49,873
May	6,000	43.8	0.62	2,444	46,317
June	6,000	38.1	0.77	2,640	42,957
July	12,000	20.3	0.62	1,133	32,090
Aug	12,000	33.2	0.62	1,853	21,942
Sept	6,000	33.2	0.8	2,390	18,333
Oct	6,000	127	0.8	9,144	21,477
TOTAL	48,000	1,001		69,477	

	Demand		Supply	
--	--------	--	--------	--

By looking at the month-end water volume column, you will have some idea about your needed volume storage. A safe estimate is to consider the largest month-end volume to be able to accommodate any fluctuation. In this example, the maximum month-end volume reached 49,873 Liters or about 50 m³. You can opt to have separate storage for all your livestock or crop irrigation, as long as the total volume equals to 50 m³ in this case.

Tips for Keeping Your Stored Water Clean and Fresh

Durability, certification for potable water use, and capacity to withstand sunlight penetration are some of the major considerations in choosing the right storage. Assuming that you have already chosen a spot for your tank and that you know your annual and monthly water requirements, you can use this information to find the perfect cistern or barrel.

Typically, your costs will cover transportation, site preparation, and the actual cost of the tank and piping system. Suppliers often quote the sizes based on the nominal capacity or the total water volume that can be stored if the tank is filled to the top. If the supplier mentions operational capacity, they are simply referring to the actual volume of water that can be stored in the tank after factoring in the overflow and fittings. In general, leave three to six inches of space just above the overflow and count three inches for water that remains in the tank after it is emptied through the tap.

Once you have procured all your containers, make sure you clean and sanitize them before using them for water storage. Newly manufactured tanks may be relatively cleaner than second-hand tanks. However, if the items were stored in the warehouse for a long time, you will still need to rinse it to avoid entrainment and mixing of dirt with the water. As much as possible, stick with food grade materials particularly for storing potable water. If you decide to buy used storage containers, choose those from food production industries. Avoid buying tanks from chemical facilities to avoid contamination.

Thorough cleaning to remove any dirt and visible grime is the first step. Once

your containers are washed and scrubbed with soap, final rinsing with hot water or wiping the inner and outer surfaces with a sanitizing solution is the next step. Use hand gloves when rinsing with hot water to avoid skin burns. A sanitizing solution, such as rubbing alcohol, will work just fine on most types of material.

Once your containers have been sanitized and dried out, you're ready to fill them with water!

Running a homestead is no easy task and providing it with enough water to function is one of the primary concerns of every homesteader - both experienced and new to the lifestyle. Once this issue has been properly resolved, you can focus your time and attention on other areas, knowing that you have overcome one of the most difficult hurdles to getting off the grid-- becoming responsible for your own water needs.

Conclusion

Thank you again for purchasing this book! I hope you have enjoyed it and are feeling prepared to handle whatever comes your way while setting up a water supply and storage system for your homestead. Now that you have explored your water supply options it is time to devise a plan as to which method or methods make sense for your situation. Once you have estimated your water needs and decided how you will gather, treat and store your water you can begin implementing your plans by starting to work on a well and setting up your storage area. It won't be long before you are up and running!

Off the grid living is difficult and involves doing almost everything "the hard way" but there is a reward. The reward is knowing that you did it all by yourself, on your own. Your water, in addition to shelter, food, and electricity are all yours by your effort alone. It's a satisfying and simple way to live, even though it may seem rough at times. The freedom and self-satisfaction inherent in this lifestyle makes a few difficult days of digging a well seem like a fair trade-off for living life on your own terms.

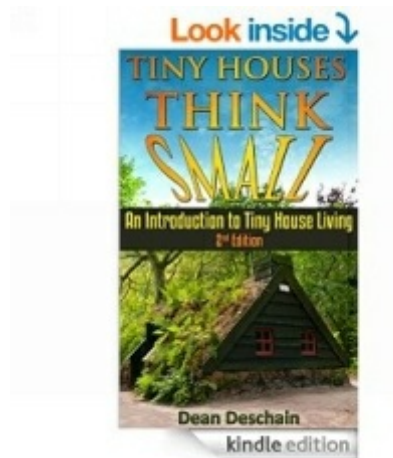
Good luck! Happy Homesteading

Mike

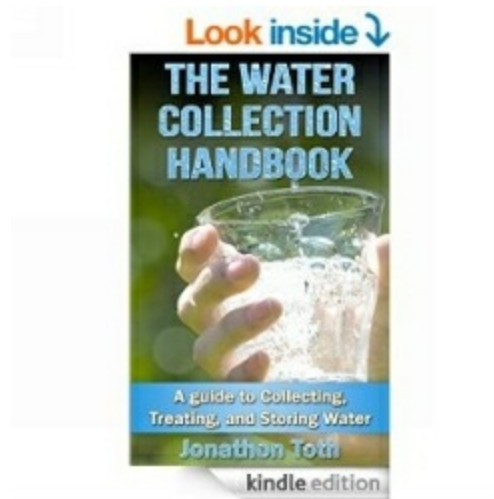


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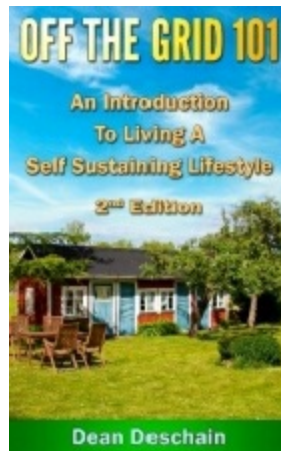
[Tiny Houses: Think Small! An Introduction to Tiny House Living \(2nd Edition\)](#)



[The Water Collection Handbook: A Guide To Collecting, Treating, and Storing Water](#)



Off the Grid 101: An Introduction to Living A Self-Sustaining Lifestyle (2nd Edition)



Mushrooms: A Beginner's Guide To Home Cultivation (2nd Edition)

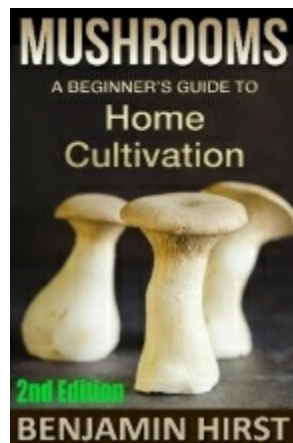


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Bonus

As an added bonus here are a couple of links to some helpful and enjoyable free water harvesting videos that discuss some of the topics and systems mentioned in this book. Enjoy!

Rainwater Harvesting by celsius thomas

https://www.youtube.com/watch?v=jhaRkQ_84Ao

DIY 1,100 Gallon Rainwater Harvesting & Collection System

By LDSPrepper

<https://www.youtube.com/watch?v=girnk6N1n9Q>

Harvesting Stream Water

by Markerbuoy – On Canada's Left Coast

<https://www.youtube.com/watch?v=F8kbVIqEaLw>

Profiting from on-farm rain water harvesting

By FarmingFutures1's channel

<https://www.youtube.com/watch?v=R8NR4bOS95g>

[1] Maximum Contaminant Level Goal: Drinking water contaminant below this level has no known health risks

[2] Maximum Contaminant Level: The highest allowable contaminant level in drinking water

[3] 75% is equal to 75 divided by 100, which is just 0.75

[\[Assumed precipitation\]](#)

[4] Based on the precipitation data obtained from your local weather station